# Richard Kublik, Ph.D.

Linars, France | rkublik@gmail.com | +33 6 07 75 94 77 | https://www.linkedin.com/in/richardkublik/

15+ years of experience | Research Engineer: Applied Mathematics, Computer Science

Ph.D. Northwestern University Dept. of Engineering Sciences and Applied Mathematics, Evanston IL, USA

I work in applied research fields where I use my expertise in mathematics and software engineering to tackle problems that no one knows how to solve. I create software and tools to improve the efficiency of teams and the algorithms they use.

## PROFESSIONAL EXPERIENCE AND TECHNICAL SKILLS

| | |
|---|---|
| February 2017 – Present<br>Research Engineer<br>Independent consultant<br>Dayton OH and Linars France | • Simulations, Algorithms, Software Prototypes<br>• Automated a 6-hour manual process<br>• Created data visualizations<br>• Clear communication with stakeholders (oral, written) |
| January 2019 – October 2025<br>Research Engineer<br>Tenet3<br>Dayton OH, USA | • Developed new capabilities in computer vision, blockchain, and data collection and analysis.<br>• Led teams of 2 to 5 engineers<br>• Cross-functional collaboration with customers, analysts, and software engineering teams |
| October 2013 – January 2017<br>Software Engineer<br>Materials Resources LLC<br>Dayton OH, USA | • Designed, developed and deployed a SaaS platform<br>• Converted 15+ command-line tools into intuitive web applications<br>• Developed new capabilities in image processing |
| March 2012 – June 2013<br>Software Engineer<br>Epic<br>Madison WI, USA | • Implemented critical features, fixing bugs, and conducted code reviews<br>• Ensured regulatory compliance by developing a critical feature for including LOINC identifiers |
| January 2011 – April 2011<br>Lecturer, Dept. of Mathematics<br>University of Texas at Austin<br>Austin TX, USA | • Taught courses in differential equations and probability<br>• 3 sections of approximately 30 students each<br>• Supervised 2 teaching assistants |
| August 2005 – December 2010<br>Graduate Research Assistant<br>Northwestern University<br>Evanston IL, USA | • Developed an efficient algorithm for simulation of electrical activity in nerve cells<br>• Design and implementation of a numerical algorithm in C++ |

## KEY SKILLS

- Computational algorithm development
- Mathematical modeling and simulation
- Software development
- Problem solving
- Applied research
- Technical writing

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Research Engineer
+15 years of experience

**KEY SKILLS**

- Computational algorithm development
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I work in applied research fields where I use my expertise in mathematics and software engineering to tackle problems that no one knows how to solve.

I create software and tools to improve the efficiency of teams and the algorithms they use.

In my career I've worked across diverse domains including computational biology, materials science, sports analytics, and supporting the United States Department of Defense. I've found that no matter the domain, the fundamental need for mathematics, problem solving skills, and computer science remain the same. I've applied these skills to help companies calibrate their laser targeting system, simulate and visualize activity in the brain, and facilitate research that improves the performance and longevity of jet engines.

I am currently interested in opportunities within engineering and research firms that need help advancing the algorithms and software that support their core capabilities.

**TECHNICAL SKILLS**

Python, Pandas, NumPy, SciPy, Scikit-learn, Scikit-image, openCV
Matplotlib, Plotly, Seaborn
JavaScript, React.js
SQL, R, C++
git, GitHub, GitLab
AWS: lambda, SNS, SQS, RDS, EC2, S3, CloudFormation

**EDUCATION**

**Northwestern University**, *Ph.D. Engineering Sciences and Applied Mathematics* — Evanston IL, USA
**University of British Columbia**, *M.Sc. Applied Mathematics* — Vancouver BC, Canada
**University of Alberta**, *B.Sc.* (Hons) Mathematics — Edmonton AB, Canada
**The King's University College**, *B.A. Music* — Edmonton AB, Canada

**PROFESSIONAL EXPERIENCE**

**Independent consultant** **Dayton OH, USA et Linars France**
**Research Engineer** **February 2017 – Present**
*I work with engineering and scientific research organizations to solve their hard problems and turn fuzzy ideas into concrete products and solutions. I typically work with companies of 50 employees and up to $12M in annual revenue.*

- Translated academic research paper into a functional algorithm to infer 3D brain activity from surface electrode measurements.
- Developed calibration procedure to facilitate laser targeting at distances over 1600m.
- Automated a manual 6 hour process to identify NFL (American football) player positions from their location on the field using tracking data.
- Created an interactive visualization tool for NFL player tracking data using plotly.js. This visualization identifies each player and their position, highlights the trajectories traveled during the action, and animates the activity captured by the NFL player tracking system.
- Contributed technical content for funded research proposals.

**Tenet3** **Dayton OH, USA**
**Research Engineer** **January 2019 – October 2025**
*Tenet3 is a software and services company that supports the United States Department of Defense. It has 50 employees and generates annual revenue of US$12 million.*

- Created a computer vision system to extract system models from schematic diagrams, thus enabling rapid modeling of existing systems.
- Automated a manual centralized blockchain system by creating a scalable web application using AWS resources.
- Led a multidisciplinary team of 5 people to collect, organize and analyze over 2 TB of text data, helping the US government client to better understand the organization that produced the documents.
- Led teams of 2 to 5 engineers to develop new capabilities in data collection and storage, distributed computing, machine learning, natural language processing, data visualization and interactive dashboards.
- Contributed to business development by creating training videos and exercises for a digital materiel management course offered by the Air Force Institute of Technology. I mentored the students of the first cohort (8 professionals from the Department of Defense) and got them excited to use MeTRA, the company's flagship software.

**Materials Resources LLC** **Dayton OH, USA**
**Software Engineer** **October 2013 – January 2017**
*MRL is a materials science research startup specializing in the acquisition and analysis of data from electron backscatter microscopy. Its clients and collaborators include the US Air Force Research Lab and GE Aerospace. During my time there, the company had between three and five employees.*

- Designed, developed and deployed a SaaS platform of data collection and analysis tools in materials science during the first six months; local deployment and using AWS cloud resources.
- Converted 15+ command-line tools into intuitive web applications, accessible to non-experts.
- Created a NoSQL database using MongoDB to store raw data and analysis results relating to more than 500 samples.

- Created new capabilities and analysis tools using computer vision and image processing techniques

**Epic** **Madison WI, USA**
**Software Engineer** **March 2012 – June 2013**
*Epic is a healthcare software company that provides electronic health record solutions used for over 78% of patients in the United States and over 3% of patients worldwide. The company has over 10,000 employees. I was one of 20 developers on the "Beaker" Clinical Laboratory team.*

- Implemented critical features, bug fixes, and conducted code reviews.
- Collaborated with the writing team to document fixes and features for clients.
- Developed a critical feature for coding LOINC identifiers to ensure the software maintained regulatory compliance

**University of Texas at Austin** **Austin TX, USA**
**Lecturer in Mathematics** **January 2011 – April 2011**
*UT Austin is a top-25 school in the USA. I taught undergraduate courses in probability and differential equations during a temporary appointment.*

- Taught 3 courses of approximately 30 students each and supervised 2 teaching assistants.

**Northwestern University** Dept of Engineering Sciences and Applied Mathematics **Evanston IL, USA**
**Applied Mathematics Thesis** **August 2005 – December 2010**
*Northwestern is a top-20 school in the USA with fewer than 10 students admitted to the ESAM program each year. I conducted independent research in computational algorithms for large scale simulations of brain activity.*

- Developed an efficient algorithm for simulation of electrical activity in nerve cells.
- Implemented algorithm using C++ to validate and demonstrate the efficiency gains.